

1940 ▶ 1945

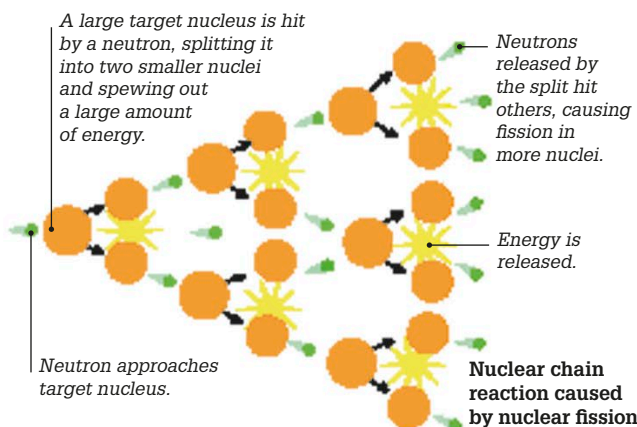
“The release of atomic energy on a large scale would be only a matter of time.”

Enrico Fermi, on the success of Chicago Pile-1

1942

Experimental nuclear reactor

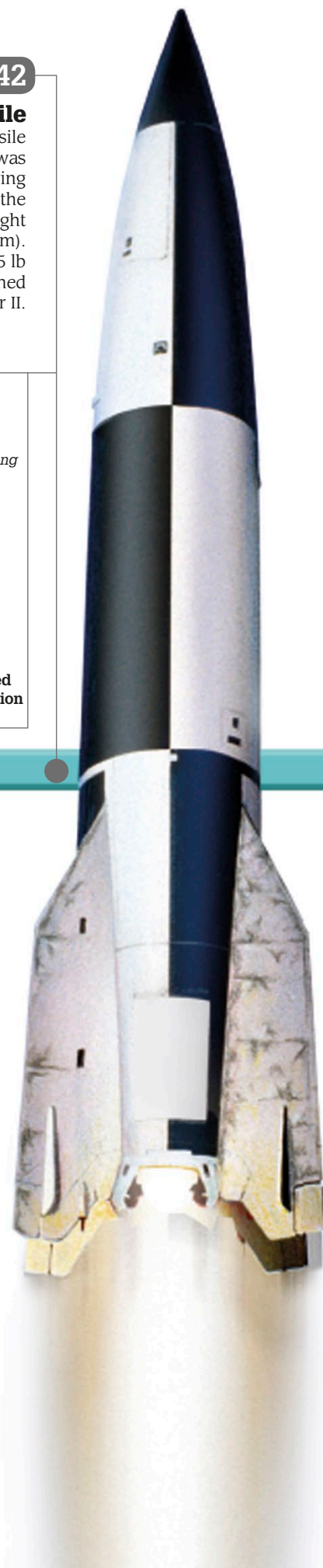
The first nuclear reactor, Chicago Pile-1, ran a nuclear chain reaction for the first time, using uranium as a fuel. Italian physicist Enrico Fermi led the team at the University of Chicago that built the reactor. When the nuclei of uranium atoms were split by nuclear fission (see p.169) in the reactor, they released energy and neutrons, which could then split more nuclei, creating a nuclear chain reaction. This would lead, eventually, to developing nuclear power stations.



1942

V2 missile

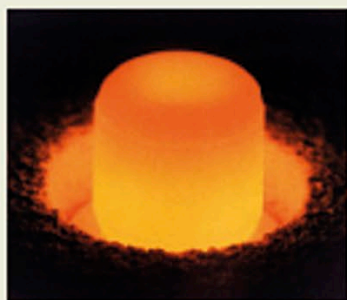
The launch of the world's first missile powered by a liquid-fueled rocket engine was tested at Peenemünde, Germany. Following three failed launches, the fourth test saw the 46-ft- (14-m-) tall rocket reach a height of more than 278,870 ft (85,000 m). Over 3,000 V2s, each carrying 1,785 lb (910 kg) of explosives, were launched on enemy targets in World War II.



1940

1940

DISCOVERY OF PLUTONIUM

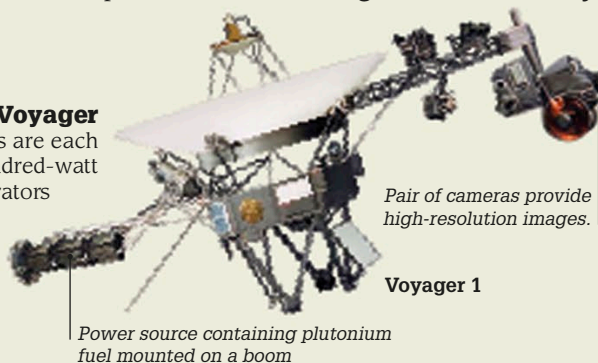


Radioactive plutonium-238 (a form of plutonium) glowing red hot

The element plutonium was discovered by American chemists Glenn Seaborg, Edwin McMillan, Joseph Kennedy, and Arthur Wahl at the University of California in 1940. Named after the dwarf planet, Pluto, this element found use in nuclear weapons—just 3.5 oz (100 g) of plutonium can produce an explosion equal to 2,204 tons (2,000 metric tons) of TNT. It is also used as a fuel in nuclear power reactors that generate electricity.

Powering Voyager

The two Voyager space probes are each powered by three multi-hundred-watt radioisotope thermoelectric generators (MHW RTGs), which used the heat from the decay of plutonium-238 fuel to generate electricity for power. Voyager 1 is now the most distant artificial object in space from Earth.



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